

# Ολοκληρώματα

## ΤΥΠΟΛΟΓΙΟ

### ■ Πολυωνυμικές συναρτήσεις

1.  $\int c \, dx = cx + c$
2.  $\int x \, dx = \frac{x^2}{2} + c$
3.  $\int x^\nu \, dx = \frac{x^{\nu+1}}{\nu+1} + c$
4.  $\int (\beta + ax) \, dx = \beta x + \frac{ax^2}{2} + c$
5.  $\int (a+x)^2 \, dx = a^2x + ax^2 + \frac{x^3}{3} + c$
6.  $\int (a+x)^\nu \, dx = \frac{(a+x)^{\nu+1}}{\nu+1} + c$

### ■ Ρητές συναρτήσεις

7.  $\int \frac{1}{x} \, dx = \log(x) + c$
8.  $\int \frac{1}{\beta + ax} \, dx = \frac{\log(\beta + ax)}{a} + c$
9.  $\int \frac{1}{(\beta + ax)^2} \, dx = -\frac{1}{a(\beta + ax)} + c$
10.  $\int \frac{1}{(\beta + x)(a + x)} \, dx = \frac{-\log(\beta + x) + \log(a + x)}{\beta - a} + c$
11.  $\int \frac{x}{\beta + ax} \, dx = \frac{-\beta \log(\beta + ax) + ax}{a^2} + c$
12.  $\int \frac{1}{x^2 + 1} \, dx = \operatorname{atan}(x) + c$
13.  $\int \frac{1}{a^2 + x^2} \, dx = \frac{\operatorname{atan}\left(\frac{x}{a}\right)}{a} + c$
14.  $\int \frac{x}{a^2 + x^2} \, dx = \frac{\log(a^2 + x^2)}{2} + c$
15.  $\int \frac{x^2}{a^2 + x^2} \, dx = -a \operatorname{atan}\left(\frac{x}{a}\right) + x + c$
16.  $\int \frac{1}{x^2 - 1} \, dx = \frac{\log(x - 1)}{2} - \frac{\log(x + 1)}{2} + c$
17.  $\int \frac{1}{-a^2 + x^2} \, dx = \frac{\log(-a + x) - \log(a + x)}{2a} + c$
18.  $\int \frac{x^3}{x^2 - 1} \, dx = \frac{x^2}{2} + \frac{\log(x^2 - 1)}{2} + c$
19.  $\int \sqrt{x} \, dx = \frac{2x^{\frac{3}{2}}}{3} + c$
20.  $\int \sqrt{1 - x^2} \, dx = \frac{x\sqrt{1 - x^2}}{2} + \frac{\operatorname{asin}(x)}{2} + c$
21.  $\int \frac{1}{\sqrt{\beta + ax}} \, dx = \frac{2\sqrt{\beta + ax}}{a} + c$

### ■ Τριγωνομετρικές συναρτήσεις

22.  $\int \sin(x) \, dx = -\cos(x) + c$
23.  $\int \cos(x) \, dx = \sin(x) + c$
24.  $\int \tan(x) \, dx = -\log(\cos(x)) + c$
25.  $\int \cot(x) \, dx = \log(\sin(x)) + c$
26.  $\int \frac{1}{\cos^2(x)} \, dx = \tan(x) + c$
27.  $\int \frac{1}{\sin^2(x)} \, dx = -\frac{1}{\tan(x)} + c$
28.  $\int \sinh(x) \, dx = \cosh(x) + c$
29.  $\int \cosh(x) \, dx = \sinh(x) + c$
30.  $\int \tanh(x) \, dx = x - \log(\tanh(x) + 1) + c$

31.  $\int \coth(x) \, dx = x - \log(\tanh(x) + 1) + \log(\tanh(x)) + c$
32.  $\int \sin(ax) \, dx = -\frac{\cos(ax)}{a} + c$
33.  $\int \cos(ax) \, dx = \frac{\sin(ax)}{a} + c$
34.  $\int \tan(ax) \, dx = -\frac{\log(\cos(ax))}{a} + c$
35.  $\int \cot(ax) \, dx = \frac{\log(\sin(ax))}{a} + c$
36.  $\int \sin^2(x) \, dx = \frac{x}{2} - \frac{\sin(2x)}{4} + c$
37.  $\int \cos^2(x) \, dx = \frac{x}{2} + \frac{\sin(2x)}{4} + c$
38.  $\int \sin(x) \cos(x) \, dx = \frac{\sin^2(x)}{2} + c$
39.  $\int \sinh(x) \cosh(x) \, dx = \frac{\cosh^2(x)}{2} + c$
40.  $\int \frac{1}{\sin(x)} \, dx = \frac{\log(\cos(x) - 1)}{2} - \frac{\log(\cos(x) + 1)}{2} + c$
41.  $\int \frac{1}{\cos(x)} \, dx = -\frac{\log(\sin(x) - 1)}{2} + \frac{\log(\sin(x) + 1)}{2} + c$
42.  $\int \frac{1}{\sin^2(x)} \, dx = -\frac{1}{\tan(x)} + c$
43.  $\int \frac{1}{\cos^2(x)} \, dx = \tan(x) + c$
44.  $\int x \sin(x) \, dx = -x \cos(x) + \sin(x) + c$

### ■ Εκθετικές - Λογαριθμικές συναρτήσεις

45.  $\int e^x \, dx = e^x + c$
46.  $\int e^{ax} \, dx = \frac{e^{ax}}{a} + c$
47.  $\int \log(x) \, dx = x(\log(x) - 1) + c$
48.  $\int \log(ax) \, dx = x(\log(ax) - 1) + c$
49.  $\int e^{\beta + ax} \, dx = \frac{e^{\beta + ax}}{a} + c$
50.  $\int xe^x \, dx = (x - 1)e^x + c$
51.  $\int x^2 e^x \, dx = (x^2 - 2x + 2)e^x + c$
52.  $\int x \log(x) \, dx = \frac{x^2(2\log(x) - 1)}{4} + c$
53.  $\int x^2 \log(x) \, dx = \frac{x^3(3\log(x) - 1)}{9} + c$
54.  $\int \frac{e^x}{x} \, dx = \operatorname{Ei}(x) + c$
55.  $\int \frac{\log(x)}{x} \, dx = \frac{\log(x)^2}{2} + c$
56.  $\int e^x \sin(x) \, dx = -\frac{\sqrt{2}e^x \cos\left(x + \frac{\pi}{4}\right)}{2} + c$
57.  $\int e^x \cos(x) \, dx = \frac{\sqrt{2}e^x \sin\left(x + \frac{\pi}{4}\right)}{2} + c$